



AUGS/AGSR DIVISION
FIRST SEMESTER 2026-27
Course Handout (Part II)

Date: 4/08/2026

In addition to Part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : ME F211/MF F211
Course Name : Mechanics of Solids
Instructor-in-charge : Prof. Sharad Shrivastava
Lecture Instructor(L2) : Prof. Kushagra Tiwari
Tutorial Instructor : Prof. Sharad Shrivastava, Prof. J.S Rathore, Prof. Sachin Belgamwar, Prof. Anuj Sharma, Prof. Kushagra Tiwari

Introduction:

Fundamental principles of mechanics, introduction of mechanics of deformable bodies, force and moment transmitted by slender members (2D truss, frame, beam), stress and strain, stress strain-temperature relations, torsion of circular member, stress and deflection due to bending, buckling of columns, theories of failure.

1. Scope and Objective:

- | Determination of strength, deformation and stability of structural and machine elements.
- | Understand the material properties and Idealization of stress-strain curves.
- | Understand different loading conditions and to analyse the results.
- | Understand combine load conditions on a body and to analyse the results
- | Understand the failure theories.

At the end of the course the student will be in a position to design and analyze simple structural elements, which involve calculation of stress, strain and deformation. This is an essential feature in any design process.

2. Text Book:

1. Crandall, Dahl and Lardner, An Introduction to Mechanics of Solids, McGraw-Hill International edition, 1978.

3. Reference Books:

1. Mechanics of Materials, Gere and Timshenko. Latest Edition.
2. Mechanics of Materials; F. P. Beer, E. R. Johnston and J. T. DeWolf, Third Edition, 2002, McGraw-Hill International Edition.
3. Introduction to Solid Mechanics by I. H. Shames, 2nd Edition, 1980, Prentice Hall of India Private Ltd. New Delhi.
4. Engineering Mechanics of Solids by E. P. Popov, 2nd Edition, PHI, New Delhi.





5. Course Plan

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1.Fundamental principles of mechanics	L1.1. Introduction, principles of mechanics, concept of force & moment, equilibrium conditions	TB Chapter 1	Fundamental principles of mechanics and to an exposition of the requirements of equilibrium
	L 1.2. concept of two & three force members, free body diagram, friction, trusses	TB Chapter 1	
	T1.1. Practice Problems	TB/RB	
2.Introduction to mechanics of deformable bodies	L2.1. Analysis of deformable bodies, uniaxial loading & deformation	TB Chapter 2	To attack problems of applied mechanics by applying three steps. The analysis of truss for both statically indeterminate and determinate conditions will be understood
	L2.2. Statical Determinate Truss	TB Chapter 2	
	L2.3. Problems on trusses	TB Chapter 2	
	L2.4. hoop stresses in thin cylindrical shells	TB Chapter 2	
	L2.5. Castiglione's Theorem	TB Chapter 2	
	L2.6.Problems	TB Chapter 2	
	T2. Problems on truss structure	TB/RB	
3. Forces & moments transmitted by slender members	L3.1. Introduction of forces & moments acting on a section of a member	TB Chapter 3	In this the study of forces and equilibrium requirements will be applied to slender members. The importance of shear force and bending moment variation along the length of slender members will be understood
	L3.2.Shear force and BM diagram	TB Chapter 3	
	L3.3 Problems	TB Chapter 3	
	L3.4 Distributed loads & resultant of distributed loads, Differential equilibrium approach, Singularity functions	TB Chapter 3	
	L3.5. Differential equilibrium approach, Singularity functions	TB Chapter 3	
	T3. Problems	TB/RB	
4. Stress & Strain	L4.1. Introduction, stress, plane stress, equilibrium of an element in plane stress,	TB Chapter 4	The investigation of three basic





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	L4.2. Mohr circle representation of a plane stress, general state of stress	TB Chapter 4	principles will be applied to the localized behavior of the materials at a point within a deformable body. Concept of stress and strain and its variation at a point in different planes will be understood.
	L4.3 Problems	TB Chapter 4	
	L4.4 Analysis of deformations, strain components, relation between strain & displacement	TB Chapter 4	
	L4.5. Strain component associated with arbitrary set of axis, Mohr circle representation of plane strain, general state of strain	TB Chapter 4	
	L4.6.Problems	TB Chapter 4	
	T4 Problems	TB/RB	
5. Stress-Strain-Temperature relations	L5.1. Introduction, tensile test, idealization of stress strain curve,	TB Chapter 5	The relation between stress and strain will be studied. Also their relation with temperature will be analysed. Different material properties and yielding criterion for ductile materials will be understood.
	L5.2. elastic stress strain relation, Thermal strain,	TB Chapter 5	
	L5.3. complete equations of elasticity, strain energy in a elastic body	TB Chapter 5	
	L5.4. Theory of failure	TB Chapter 5	
	L5.5 Problems	TB Chapter 5	
	T5.Problems	TB/RB	
6. Torsion	L6.1. Introduction, geometry of deformation of a twisted circular shaft, stress strain relations, ,	TB Chapter 6	In this chapter student will apply the fundamental principles to consider the problem of twisting. The stress developed due to torsion will be analyzed and corresponding
	L6.2 equilibrium requirements, stresses & deformations in twisted elastic circular shaft	TB Chapter 6	
	L6.3. torsion of elastic hollow circular shaft,	TB Chapter 6	
	L6.4, combined stresses, strain energy due to torsion, yielding in torsion	TB Chapter 6	
	L6.5.Problems	TB Chapter 6	
	T6. Problems	TB/RB	deformation will also be understood.



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7.Stresses due to bending	L7.1.Introduction, deformation in pure bending, stress-strain relations, equilibrium requirements,	TB Chapter 7	Here the slender member subjected to transverse loading would be analysed. The stresses and deformation would be analysed .
	L7.2 stresses & deformations in pure bending.	TB Chapter 7	
	L7.3. Stresses due to shear force and bending moment,	TB Chapter 7	
	L7.4. combined stresses, strain energy due to bending, yielding in bending	TB Chapter 7	
	L7.5.Problems	TB Chapter 7	
	T7.Problems	TB/RB	
8. Deflections due to bending	L8.1.Introduction, moment-curvature-relations, integration of moment-curvature relations, Numerical problems	TB Chapter 8	The deflection of slender members which transmit bending moments would be analysed, which would be helpful while designing high speed machinery with close tolerances, leaf springs etc.
	L8.2. superposition, Load-deflection	TB Chapter 8	
	L8.3 differential equation, Energy Methods, Problems	TB Chapter 8	
	T8.Problems	TB/RB	
9. buckling of columns	L9.1 Introduction, elastic stability, examples of instability,	TB Chapter 9	In this chapter student will analyse the body when it is deviated from equilibrium conditions.The analysis of columns would be done to predict the stability of structure.
	L9.2 elastic stability of flexible columns	TB Chapter 9	
	L9.3Problems	TB Chapter 9	



6. Evaluation Scheme:

EC No.	Evaluation Component	Duration	Weightage	Date, Time & Venue	Nature of Component
2	MID SEM	90min	30% (60marks)	10/10 AN	CB
3	TUTORIAL	50 min	20% (40 marks)	Will be announced in class	OB
4	Surprise test in Tutorial		10% (20 marks)	Surprise	CB/OB
4	Compre. Exam.	180min	40%(80marks)	16/12 FN	CB/OB

7. **Tutorial :** Will be announced in the lecture class. **There will be no makeup in any circumstances for evaluative tutorials/quizzes. There will be total of 5 evaluative tutorials out of which best 4 would be considered.**

Attendance Policy: *Each student is expected to attend all classes, tutorials, and practical sessions and to participate in all evaluation components. A 10% weight is assigned to assessment linked to attendance and participation. To this end, a small surprise quiz will be conducted in the tutorial class. The scores in surprise quizzes linked to attendance and participation will be considered in the final score. This score will be considered only for students with a minimum attendance of 50% in the lecture and tutorial classes separately. For students with attendance less than 50% in either the lecture or tutorial class, zero marks will be assigned, irrespective of the scores in the surprise quizzes attempted.*

8. **Notices:** All notices concerning the course will be uploaded on the **Nalanda** portal only.
<https://nalanda.bits-pilani.ac.in/>

The passing marks for this course will be determined based on one of the two criteria:
30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course.

9. **Make-up:**

- Make-up request for mid-semester examination is to be submitted 7 days before the exam (or within 72 hrs in case of medical emergency). For comprehensive examination the request is to be made to AUGSD as per its guidelines.
 - It is mandatory to submit the necessary supporting documents (such as: listed below) along with make-up request:
 - Letter of invitation/selection for a national/international event where the applicant is representing BITS Pilani.
 - Proof of hospitalization, medical certificate, prescription, medicine purchase bill , etc.
 - The decision of granting make-up to the applicant is taken mutually by the team of course instructors and would be final in all such matters.
 - **There is no makeup for the tutorial and quiz.**
 - Make-up in mid sem and compre exams only will be granted only to genuine cases.
10. **Online Consultation Hour:** Will be announced by instructors individually during the tutorial class.

Instructor-in-charge ME F211/ MF F211



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